

Cost Benefit Analysis

Cost-benefit analysis (CBA) is a systematic approach used to evaluate the financial implications of a decision or project by comparing its projected costs and benefits.

The goal is to determine if the potential benefits outweigh the costs, helping decision-makers make informed choices.

Cost-benefit analysis formulas

There are three formulas to help perform a basic cost-benefit analysis: The present value (PV) formula, the net present value (NPV) formula, and the cost-benefit analysis (CBA) ratio.

To calculate the CBA ratio, you must first use the PV formula to find the present value of the costs and the benefits of the project that's being analyzed.

$$\text{Present Value (PV)} = \text{FV} / (1+r)^n$$

Where:

- PV = Present Value
- FV = Future Value
- r = Discount Rate
- n = Number of periods

Apply this formula to get the PV of future costs and PV of future benefits.

Then, insert those values into the NPV formula and CBA ratio, as seen below:

$$\text{NPV} = \text{PV of Benefits} - \text{PV of Costs}$$

$$\text{CBA ratio} = \text{PV of Benefits} / \text{PV of Costs}$$

If the NPV is a positive number and the CBA ratio is greater than 1, that means the present value of the benefits outweighs the present value of the costs. Therefore, the project being analyzed is financially feasible.

Example 1 - Cost-benefit analysis (CBA) example with formulas

Example 1: Let's consider a hypothetical example of a CBA for a company deciding whether to invest in new manufacturing equipment:

Project: Purchase of new manufacturing equipment

- Cost of equipment: \$500,000
- Expected useful life: 5 years
- Annual maintenance cost: \$20,000
- Expected annual increase in production: 10,000 units
- Selling price per unit: \$50
- Variable cost per unit: \$30

- Discount rate: 10%

Step 1: Calculate potential annual financial benefits

Annual gross profit = 10,000 units x (\$50 selling price - \$30 cost) = \$200,000

Annual net profit = \$200,000 gross profit - \$20,000 maintenance = \$180,000

The potential annual financial benefit of new manufacturing equipment is \$180,000 per year.

Step 2: Calculate present value of financial benefits

To calculate the total present value (PV) for the potential financial benefits, we use the expected useful life of the new equipment (five years) as our 'r' value in the PV formula and calculate PV for each year.

Year 1 PV of benefits = \$180k / (1+10%)¹ = 163,636.36

Year 2 PV of benefits = \$180k / (1+10%)² = 148,760.24

Year 3 PV of benefits = \$180k / (1+10%)³ = 135,164.73

Year 4 PV of benefits = \$180k / (1+10%)⁴ = 122,084.24

Year 5 PV of benefits = \$180k / (1+10%)⁵ = 111,024.36

Finally, we sum the PV of each of the next five years to get the total PV of the potential financial benefits of this project.

Total PV of Benefits = \$680,669.93

Step 3: Calculate present value of costs

Next, we need to calculate the total present value of costs by adding the upfront cost (\$500,000) to the present value of annual maintenance costs. To calculate the present value of annual maintenance costs, we'll use the PV formula again.

Year 1 PV of costs = \$20k / (1+10%)¹ = 18,181.82

Year 2 PV of costs = \$20k / (1+10%)² = 16,528.93

Year 3 PV of costs = \$20k / (1+10%)³ = 15,029.77

Year 4 PV of costs = \$20k / (1+10%)⁴ = 13,658.69

Year 5 PV of costs = \$20k / (1+10%)⁵ = 12,345.68

Now we sum up these present values for the PV of maintenance costs.

PV of Annual Maintenance Costs = \$75,744.89

Finally, we can calculate the total present value of costs, which includes both the initial cost of equipment (\$500,000) and the present value of maintenance costs (\$75,744.89).

$$\text{Total PV of Costs} = 500,000 + 75,744.89 = \$575,744.89$$

Step 4: Calculate NPV and CBA ratio

Finally, we'll insert our calculated values into the NPV formula and CBA ratio.

$$\text{NPV} = \text{PV of Benefits} - \text{PV of Costs}$$

$$\text{Net Present Value (NPV)} = \$680,669.93 - \$575,744.89 = \$104,925.04$$

$$\text{Since, CBA ratio} = \text{PV of Benefits} / \text{PV of Costs}$$

$$\text{CBA Ratio} = \$680,669.93 / \$575,744.89 = 1.18$$

In this example, the positive NPV and benefit-cost ratio greater than 1 suggest that the investment in new equipment is financially viable.

Software Engineering CBA

A software engineering Cost-Benefit Analysis (CBA) evaluates whether the financial benefits of a project (e.g., increased revenue, time savings) exceed its costs (development, maintenance, and training) over a specific time.

For example, a company upgrading to a new CRM, costing \$100,000 to implement, might realize \$150,000 in annual productivity gains, yielding a positive return within a year.

Key Components of Software CBA

- **Costs:** Licensing fees, hardware/infrastructure, labor, training, and operational downtime.
- **Benefits:** Increased sales revenue, decreased operational costs due to automation, enhanced security, and improved employee productivity.

Example 2 - Developing a Custom SaaS Tool

A company is deciding whether to build a custom internal tool or buy a third-party solution.

- **Costs (Development & Operation):**
 - **Development Labor:** \$100,000 (Engineers/Designers).
 - **Infrastructure:** \$10,000 (Cloud servers).
 - **Maintenance:** \$5,000/year.
- **Benefits (Projected over 3 years):**
 - **Time Savings:** \$60,000/year (Automated reporting reduces staff hours).
 - **Increased Revenue:** \$20,000/year (Enhanced functionality).

In the example above, the total 3-year benefit (\$240,000) far outweighs the estimated 3-year cost (\$125,000), making it a sound investment.

Analysis Techniques

- Cost-Benefit Ratio (CBR) = Total Benefits / Total Costs.
- Net Present Value (NPV) = PV of Benefits - PV of Costs
- Return on Investment (ROI) = ((Total Benefits – Total Costs) / Total Costs) x 100
- Payback Period = Initial Investment (Project Cost) / Annual Net Cash Flow (Benefits)

Example 3 - Developing a Customer Support Chatbot

- **Step 1: Define Project Scope & Goals**

- *Example:* Build an AI Chatbot to handle 50% of customer support tickets, reducing manual agent workload over 3 years.

- **Step 2: Identify Costs and Benefits**

- **Costs:** Development salaries (\$100k), API fees (\$10k/year), training (\$5k).
- **Benefits:** Reduced agent labor (\$60k/year), increased sales from fast support (\$20k/year).

- **Step 3: Assign Monetary Value (1-Year Projection)**

- **Total Costs:** $\$110k + \$10k + \$5k = \$115,000$.
- **Total Benefits:** $\$60k + \$20k = \$80,000$.

- **Step 4: Calculate Net Benefit / Present Value**

- *Note:* Use a discount rate to factor in the time value of money, typically calculated over 3–5 years.
- Year 1 Net Benefit = $\$80,000 - \$115,000 = \$35,000$ (Loss).
- Assuming constant benefits, Year 2 and 3 show profit.

- **Step 5: Compare Costs vs. Benefits & Analyze Risks**

- **Benefit-Cost Ratio:** Total Benefits / Total Costs.
- *Example:* If 3-year total benefits = \$240k and costs = \$135k, CBA ratio = $240 / 135 = 1.77$.
A ratio >1 indicates a good investment.
- **Risks:** Low user adoption, high maintenance costs.

- **Step 6: Make a Decision**

- If the NPV is positive over the project lifespan and the risks are manageable, proceed with development.

Key considerations include tangible benefits (e.g., reduced labor) and intangible benefits (e.g., improved reputation).